Higher resolution, longer particle shock histories

Measurement	Actual recorded result
Grid / particles	128^3 / 8,192
End time / generations	1 / 6033
Complete / partial / open	367,074 / 0 / 999
Complete episodes / particle	44.8088
Final volume rms Mach	0.8086
Final supersonic volume fraction	20.40%
Validated per-step observations	49,430,528
Native run wall time	126.5 minutes

Experiment

Periodic unit cube, ideal Newtonian gas gamma=1.4, native RK2/PLM/HLLC and CFL0.30. Initial velocity rms Mach5, continuous native edot50 forcing, the same seeds and8,192 mass-flux MC tracers as the earlier64-cubed run. Detector width8 and8ghost cells preserve the physical endpoint half-span1/16box. This run evolves to t=1, compared with the earlier t=0.3. No production detector, particle or forcing machinery changed.

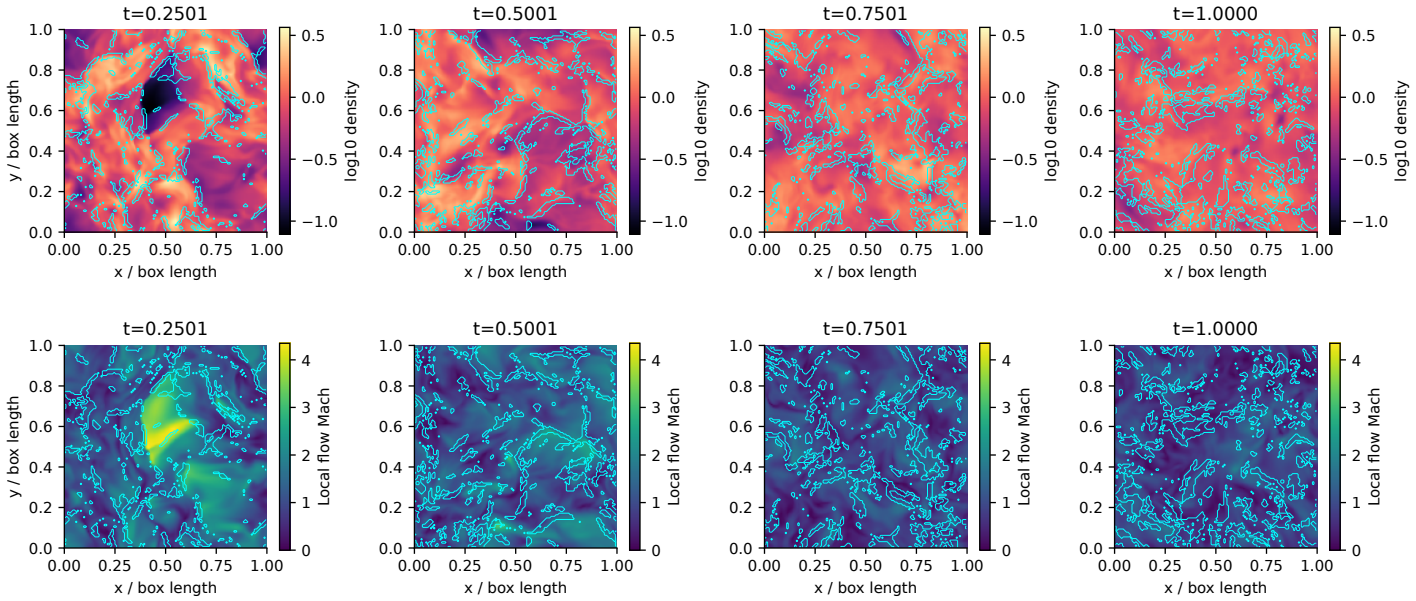
What passed

Exact frozen ShockIt mask/strength parity for new width8 fixtures, including an independent planar Mach3 jump oracle. Full per-step traces retain all particle IDs, raw shock flags, phases and precise observation clocks. Canonical event sequences are unique and contiguous; native counters agree. Positive finite checkpoint fluid states conserve periodic mass to1e-12. Same-binary zero-step checkpoint recovery matches every particle thermodynamic sample; plotted native detector slices match live particle mask/strength evidence to the unchanged1e-12 tolerance.

Limits and reproducibility

The inherited RK2 forcing thermal oracle remains a documented failure. Adiabatic heating can reduce the flow Mach despite continuous forcing. These are sampled numerical encounters, not certified physical shock counts or exact crossing times. Stationarity, ensembles, convergence, RUSTY/Frontier performance and human scientific approval remain open. The README records exact commands, source/executable/input hashes, output-cadence recovery, full trace validation and raw-data paths.

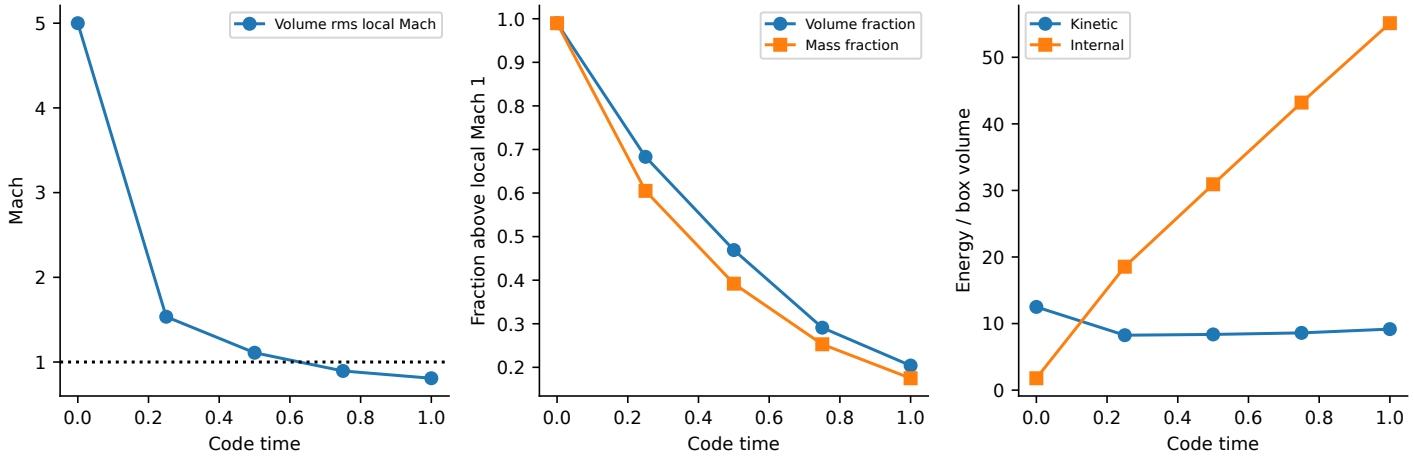
01. 128-cubed flow and detected regions



Actual z=64 cell-center slices (z=0.50390625). Top: log10 density; bottom: local speed/sound speed. Each row uses shared color limits spanning all plotted data, with no clipped tails. Cyan outlines use the native unreduced width8 detector. The central crop has full real support; masks and finite/NaN strength values agree with actual live particle samples on this plane at rtol=atol=1e-12. Checkpoint maps were recovered with the same binary and zero evolved steps. These detector masks are numerical evidence, not independently certified physical shock surfaces.

Data: 01_flow.pdf / .svg / .png; episodes.csv, evolution.csv, selected-traces.csv and report.json.

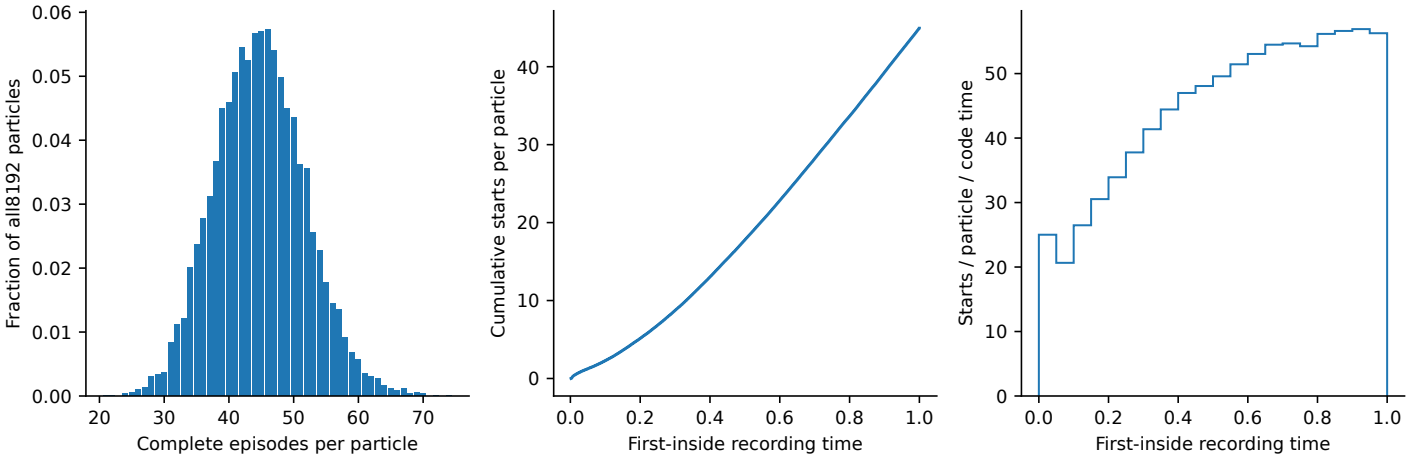
02. Flow evolution under continuous forcing



Volume rms Mach = sqrt(mean(v^2/(gamma P/rho))). The gas is adiabatic: initial Mach5 and continuous edot50 do not hold its sound speed or Mach number fixed. Lines only connect the saved checkpoint states. Actual mass stays within1e-12 of1. Forcing energy is not independently calibrated because the inherited RK2 source thermal oracle fails; no stationary regime is asserted.

Data: 02_evolution.pdf / .svg / .png; episodes.csv, evolution.csv, selected-traces.csv and report.json.

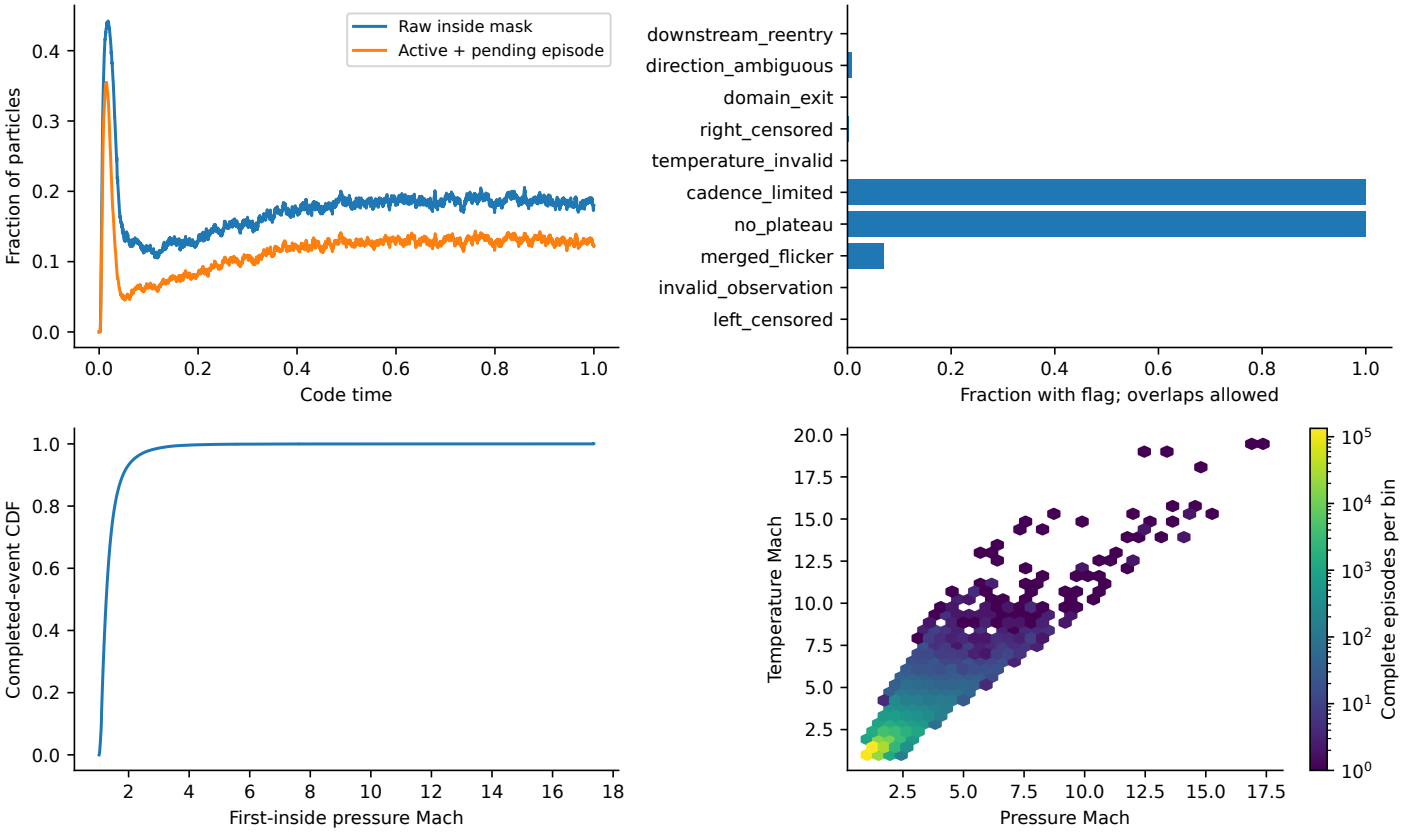
03. Accumulated particle encounter statistics



All8192 IDs are included, including zero-complete particles. Starts count every committed or terminal-open identity once at its first-inside sample. Partial/open outcomes are retained separately. Rate bins cover the stated time window; these are numerical recording rates, not exact physical shock crossing rates or independent Poisson trials.

Data: 03_encounters.pdf / .svg / .png; episodes.csv, evolution.csv, selected-traces.csv and report.json.

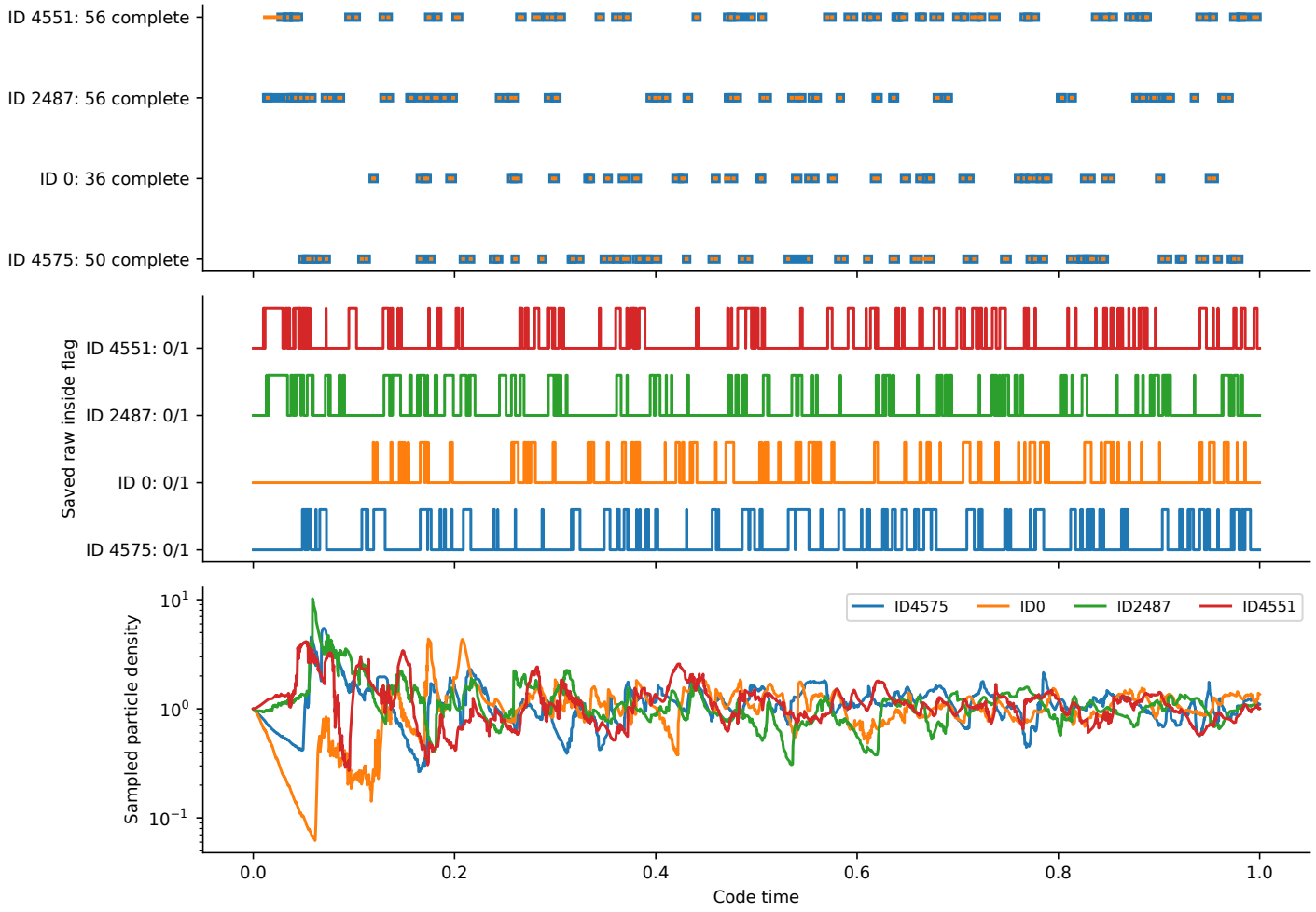
04. Shock flags, episode state and strength evidence



The full per-step trace explicitly saves each raw inside flag. Raw membership and encounter phase differ: a pending exit can be outside while the episode remains open, and a suppressed downstream reentry can be inside while its phase is outside. Both are plotted from recorded states. Suppressed reentry attempts are a separate native counter, not extra event identities. All validity bits are retained. Finite first-inside strength estimates compare two native estimators; their agreement is not a physical calibration. Either estimator can accept membership, so a pressure estimate belowmin_mach1.1 can remain present.

Data: 04_membership_strength.pdf / .svg / .png; episodes.csv, evolution.csv, selected-traces.csv and report.json.

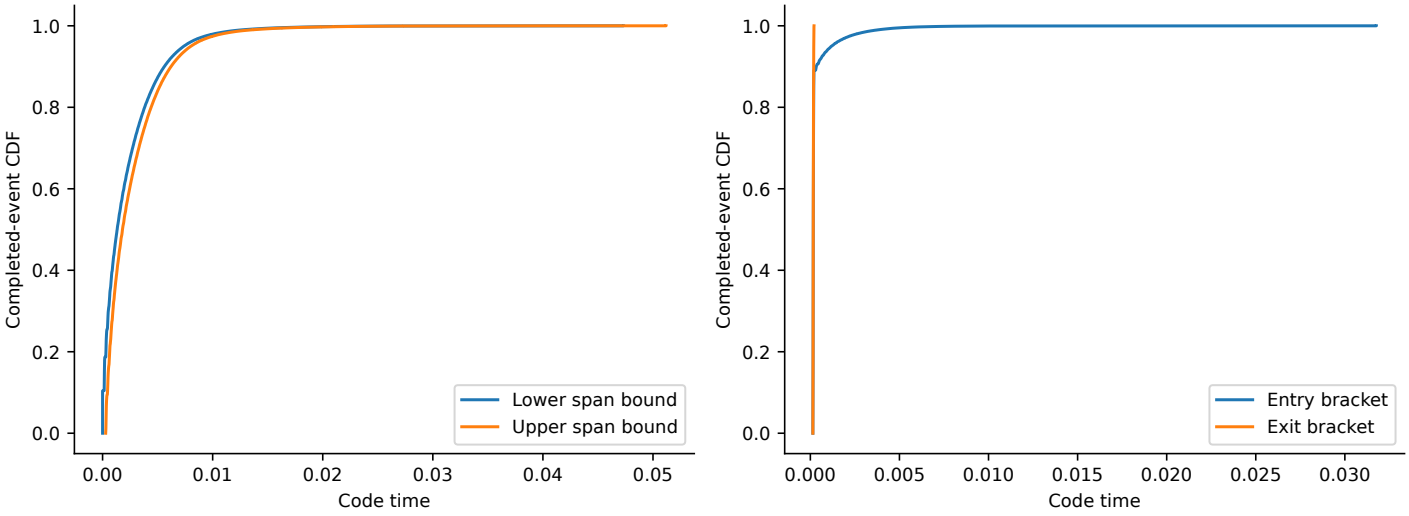
05. Actual per-step particle flags and encounter histories



Same IDs as the earlier64-cubed history illustration, with fresh128-cubed realizations. Top: blue first-to-last inside evidence, orange entry/exit brackets, red right-censored terminal marker. Middle: every saved raw0/1 inside flag, vertically offset by particle. Bottom: actual every-step density samples. Connecting samples guides the eye; it does not invent continuous trajectories or exact crossing times. Selected IDs are illustrations, not unbiased ensemble statistics.

Data: 05_particle_histories.pdf / .svg / .png; episodes.csv, evolution.csv, selected-traces.csv and report.json.

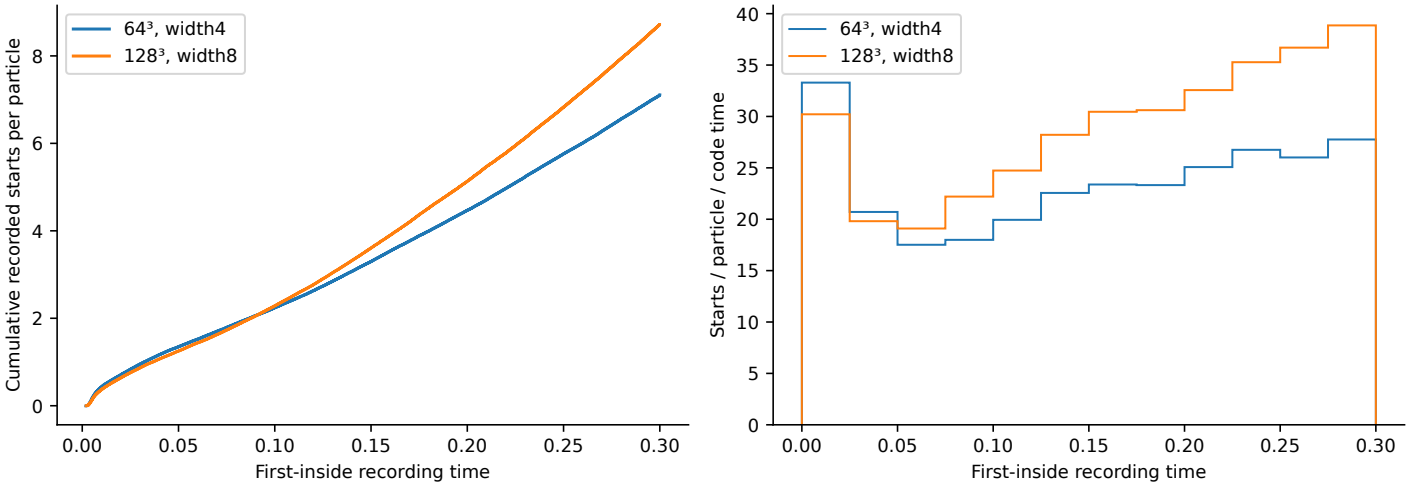
06. Timing bounds from observed samples



For complete episodes the entry bracket is[pre,first], exit bracket[last,post], and elapsed-span interval[last-first,post-pre]. No midpoint is assigned. Merged flicker may include brief outside intervals, so this span is not continuous inside residence time. Terminal-open records receive no artificial completed duration.

Data: 06_timing.pdf / .svg / .png; episodes.csv, evolution.csv, selected-traces.csv and report.json.

07. Matched detector span in the common time window



Both cases use8192 particles, CFL.30 and the same seeds/forcing inputs. Width8 at128³ matches width4 at64³ physical half-span1/16box. Only first-inside starts through the common observed endpoint are compared, including unique terminal-open identities; unequal later closure times do not bias this measure. Resolution still changes fluid evolution, MC random draws and observation cadence together. This is measured sensitivity, with no claimed converged tolerance or independent ensemble error bar.

Data: 07_resolution.pdf / .svg / .png; episodes.csv, evolution.csv, selected-traces.csv and report.json.